Rapido Ride Data Analysis

SQL Data Analyst Portfolio Project

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📌 Project Summary

This project simulates a real-world ride-hailing data analysis workflow using SQL.
The dataset represents ride booking data from Rapido's transportation platform and includes booking value, ride distance, ride time, customer and driver ratings, cancellations, and operational metrics.

The objective was to:

- Build a structured SQL database from raw Excel data
- Perform exploratory data analysis (EDA)
- Clean inconsistent and invalid ride data
- Generate business-driven insights using SQL queries

This project demonstrates practical SQL skills aligned with transportation, mobility, and operational analytics use cases.

📊 Dataset Overview

- Source: Rapido Ride Dataset (Excel File)
- Total Records: 200 Ride Bookings
- Valid Records After Cleaning: 199
- Completed Rides: 165
- Cancelled Rides: 24
- Incomplete Rides: 10
- Cities Covered: Hyderabad, Delhi, Chennai, Pune, Bengaluru

Each row represents a unique ride booking, reflecting real-world ride-hailing complexity including cancellations, incomplete rides, driver performance, and operational turnaround metrics.

🛠 Tools & Technologies Used

- MySQL
- MySQL Workbench

- SQL (Aggregations, Grouping, Filtering, CASE Statements)
- Excel Data Handling

🗄️ Database Design

Created a structured relational table with appropriate data types:

```
CREATE TABLE rapido (
Booking_ID VARCHAR(30) PRIMARY KEY,
Booking_Status VARCHAR(50),
Booking_Value DECIMAL(10,2),
Customer_ID VARCHAR(30),
Driver_ID VARCHAR(30),
Pickup_Location VARCHAR(100),
Drop_Location VARCHAR(100),
Ride_Distance DECIMAL(8,2),
Ride_Time INT,
Vehicle_Type VARCHAR(50),
Payment_Method VARCHAR(50),
Customer_Rating DECIMAL(3,1),
Driver_Rating DECIMAL(3,1),
Canceled_Rides_by_Customer INT,
Canceled_Rides_by_Driver INT,
Incomplete_Rides INT,
Canceled_Percentage DECIMAL(5,2),
V_TAT INT,
C_TAT INT
);
```

🔍 Exploratory Data Analysis (EDA)

Key exploration steps:

- Counted total ride records
- Analyzed booking status distribution
- Identified distinct vehicle types
- Evaluated payment method distribution
- Examined city-wise revenue performance
- Checked for NULL values

Observation:

- Majority of rides were successfully completed (83% completion rate)
- Bike rides dominate platform usage
- Hyderabad generated the highest total revenue
- UPI is the most preferred payment method

🧹 Data Cleaning

1️⃣ Removed Invalid Booking Value Rows

Rides where Booking_Value ≤ 0 were deleted.

```
DELETE FROM rapido
WHERE Booking_Value <= 0;
```

2️⃣ Removed Invalid Distance Rows

Rides where Ride_Distance ≤ 0 were deleted.

```
DELETE FROM rapido
WHERE Ride_Distance <= 0;
```

This step ensured data consistency and accuracy for revenue and performance analysis.

📈 Business Insights Derived

1. Highest Revenue Ride

Identified the ride with the highest booking value (₹246.90).

2. High-Revenue City

Hyderabad generated the highest total revenue among all cities.

3. Revenue by Payment Method

UPI contributed the highest share of revenue, indicating strong digital adoption.

4. Driver Performance Analysis

Multiple drivers maintained an average rating close to 5.0, indicating strong service quality.

5. Revenue Efficiency Analysis

Calculated Revenue per KM to identify highly profitable short-distance rides.

📌 Key Analytical Takeaways

✓ 83% of rides were successfully completed
✓ Hyderabad is the strongest performing market
✓ Bike is the most frequently used vehicle type
✓ UPI drives the majority of digital revenue
✓ Revenue per KM analysis highlights pricing efficiency opportunities

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💡 Business Impact Simulation

If this were a real mobility analytics role, insights from this analysis could help:

- Optimize dynamic pricing strategy
- Reduce ride cancellations
- Improve driver allocation efficiency
- Increase revenue through high-value short rides
- Improve turnaround time (TAT) performance
- Enhance customer satisfaction through rating analysis

🎯 Skills Demonstrated

- SQL Database Design
- Data Cleaning & Transformation
- Exploratory Data Analysis
- Business KPI Derivation
- Transportation & Mobility Analytics
- Analytical Thinking

🚀 Why This Project Matters

This project replicates real-world tasks performed by:

- Transportation Data Analysts
- Mobility Analysts
- Operations Analysts
- Business Intelligence Analysts

It showcases end-to-end workflow:

Raw Data → Database → Cleaning → Analysis → Business Insights

SQL QUERY AND OUTPUTS (All 24 Tasks)

1.Count Total Records

SELECT COUNT(*) AS Total_Rides

FROM rapido;

COUNT(*)
199

2.Booking Status Distribution

SELECT Booking_Status,

COUNT(*) AS Total_Rides

FROM rapido

GROUP BY Booking_Status;

Booking_Status	Total_Rides
Completed	165
Cancelled	24
Incomplete	10

3.Vehicle Type Distribution

SELECT Vehicle_Type,

COUNT(*) AS Total_Rides

FROM rapido

GROUP BY Vehicle_Type;

Vehicle_Type	Total_Rides
Bike	140
Auto	59

4.City-wise Revenue

SELECT Pickup_Location,

SUM(Booking_Value) AS Total_Revenue

FROM rapido

GROUP BY Pickup_Location

```
ORDER BY Total_Revenue DESC;
```

City	Total_Revenue (₹)
Hyderabad	6929.01
Delhi	5511.27
Chennai	5445.17
Pune	5360.46
Bengaluru	4602.14

5.Top 5 Highest Revenue Rides

```
SELECT Booking_ID,

Booking_Value

FROM rapido

ORDER BY Booking_Value DESC

LIMIT 5;
```

Booking_ID	Booking_Value (₹)
RAP20250700035	246.9
RAP20250700052	246.2
RAP20250700070	244.8
RAP20250700054	244.15
RAP20250700033	243.7

6.Revenue by Payment Method

```
SELECT Payment_Method,

SUM(Booking_Value) AS Revenue

FROM rapido

GROUP BY Payment_Method

ORDER BY Revenue DESC;
```

Payment_Method	Revenue (₹)
UPI	12336.17
Wallet	7888.37
Cash	6239.86
Card	1383.65

7.Top Drivers by Rating

```
SELECT Driver_ID,

ROUND(AVG(Driver_Rating),2) AS Avg_Rating

FROM rapido

GROUP BY Driver_ID
```

```
ORDER BY Avg_Rating DESC

LIMIT 5;
```

Driver_ID	Avg_Rating
DR_219	5.0
DR_790	5.0
DR_879	5.0
DR_573	4.9
DR_740	4.9

8.Revenue per KM (Top 5)

```
SELECT Booking_ID,

ROUND(Booking_Value / Ride_Distance,2) AS Revenue_Per_KM

FROM rapido

ORDER BY Revenue_Per_KM DESC

LIMIT 5;
```

Booking_ID	Revenue_per_KM (₹)
RAP20250700137	131.99
RAP20250700057	121.89
RAP20250700113	118.03
RAP20250700189	101.41
RAP20250700135	79.96

9.Completed Ride Count

```
SELECT COUNT(*) AS Completed_Rides

FROM rapido

WHERE Booking_Status = 'Completed';
```

Metric	Value
Completed_Rides	165

10.Cancelled Ride Count

```
SELECT COUNT(*) AS Cancelled_Rides

FROM rapido

WHERE Booking_Status = 'Cancelled';
```

Metric	Value
Cancelled_Rides	24

11.Incomplete Ride Count

```
SELECT COUNT(*) AS Incomplete_Rides
```

```
FROM rapido

WHERE Booking_Status = 'Incomplete';
```

Metric	Value
Incomplete_Rides	10

12.Completion Rate

```
SELECT

ROUND(

(SUM(CASE WHEN Booking_Status='Completed' THEN 1 ELSE 0 END)

/ COUNT(*)) * 100,2

) AS Completion_Rate_Percentage

FROM rapido;
```

Metric	Value
Completion_Rate (%)	83%

13.Total Revenue

```
SELECT SUM(Booking_Value) AS Total_Revenue

FROM rapido;
```

Metric	Value
Total_Revenue (₹)	27848.05

14.Highest Revenue City

```
SELECT Pickup_Location,

SUM(Booking_Value) AS Total_Revenue

FROM rapido

GROUP BY Pickup_Location

ORDER BY Total_Revenue DESC

LIMIT 1;
```

Metric	Value
City	Hyderabad

15.Lowest Revenue City

```
SELECT Pickup_Location,

SUM(Booking_Value) AS Total_Revenue
```

```
FROM rapido

GROUP BY Pickup_Location

ORDER BY Total_Revenue ASC

LIMIT 1;
```

Metric	Value
City	Bengaluru

16.Most Used Payment Method

```
SELECT Payment_Method,

COUNT(*) AS Usage_Count

FROM rapido

GROUP BY Payment_Method

ORDER BY Usage_Count DESC

LIMIT 1;
```

Metric	Value
Payment_Method	UPI

17.Least Used Payment Method

```
SELECT Payment_Method,

COUNT(*) AS Usage_Count

FROM rapido

GROUP BY Payment_Method

ORDER BY Usage_Count ASC

LIMIT 1;
```

Metric	Value
Payment_Method	Card

18.Average Driver Rating

```
SELECT ROUND(AVG(Driver_Rating),2) AS Avg_Driver_Rating

FROM rapido;
```

Metric	Value
Avg_Driver_Rating	4.8

19.Average Customer Rating

SELECT ROUND(AVG(Customer_Rating),2) AS Avg_Customer_Rating

FROM rapido;

Metric	Value
Avg_Customer_Rating	4.6

20.Longest Average Distance City

SELECT Pickup_Location,

ROUND(AVG(Ride_Distance),2) AS Avg_Distance

FROM rapido

GROUP BY Pickup_Location

ORDER BY Avg_Distance DESC

LIMIT 1;

Metric	Value
City	Chennai

21.Most Used Vehicle Type

SELECT Vehicle_Type,

COUNT(*) AS Usage_Count

FROM rapido

GROUP BY Vehicle_Type

ORDER BY Usage_Count DESC

LIMIT 1;

Metric	Value
Vehicle_Type	Bike

22.Least Used Vehicle Type

SELECT Vehicle_Type,

COUNT(*) AS Usage_Count

FROM rapido

GROUP BY Vehicle_Type

```
ORDER BY Usage_Count ASC

LIMIT 1;
```

Metric	Value
Vehicle_Type	Auto

23.Highest Single Ride Revenue

```
SELECT MAX(Booking_Value) AS Highest_Ride_Revenue

FROM rapido;
```

Metric	Value
Booking_Value (₹)	246.9

24.Highest Revenue per KM

```
SELECT MAX(Booking_Value / Ride_Distance) AS Highest_Revenue_Per_KM

FROM rapido;
```

Metric	Value
Revenue_per_KM (₹)	131.99